Charge-Current Continuity

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Lecture Objective

- To examine the charge-current continuity equation and a consequence thereof, namely KCL.

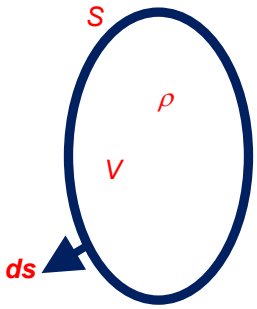
Conservation of Charge

- Physical principle: “Charge is conserved in any closed physical system.”
- Since current is the flow of charge and charge is conserved, there must be a relationship between the current flow out of a region and the rate of change of the charge contained within the region.

Conservation of Charge

- Consider a volume V bounded by a closed surface S in a homogeneous medium of permittivity ε and conductivity σ containing charge density ρ.
- The net current leaving V through S must be equal to the time rate of decrease of the total charge within V, i.e.,

$$I = -\frac{dQ_{enc}}{dt}$$



Conservation of Charge

- The net current leaving the region is given by

$$I = \oint_S \vec{J} \cdot d\vec{s}$$

- The total charge enclosed within the region is given by

$$Q = \int_V \rho_V \, dv$$

Conservation of Charge

- Hence, we have

$$\boxed{\oint_S \vec{J} \cdot d\vec{s} = -\frac{d}{dt}\int_V \rho_V \, dv}$$

net outflow of current from the region

net rate of decrease of total charge contained within the region

Continuity Equation

- Using the *divergence theorem*, we have for the LHS:

$$\oint_S \vec{J} \cdot d\vec{s} = \int_V \nabla \cdot \vec{J} \, dv$$

- For the RHS:

Becomes a partial derivative when moved inside of the integral because ρ is a function of position as well as time.

$$\frac{d}{dt} \int_V \rho_V \, dv = \int_V \frac{\partial \rho_V}{\partial t} \, dv$$

Continuity Equation

- Thus,

$$\int_V \nabla \cdot \vec{J}\, dv + \int_V \frac{\partial \rho_V}{\partial t}\, dv = 0$$

- Since the above relation must be true for *any and all regions*, we have

$$\boxed{\nabla \cdot \vec{J} + \frac{\partial \rho_V}{\partial t} = 0}$$

Continuity Equation

Kirchoff's Current Law

- For steady currents, $$\frac{\partial \rho_V}{\partial t} = 0$$

- Thus, the volume current density is a solenoidal vector:

$$\nabla \cdot \vec{J} = 0$$

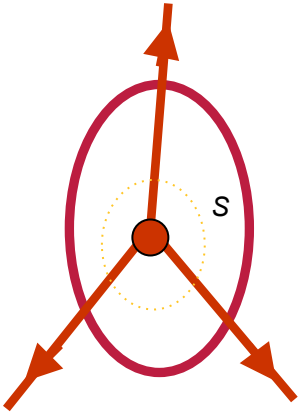
Kirchhoff's Current Law

- Since the current density vector is solenoidal, we must have

$$\oint_S \vec{J} \cdot d\vec{s} = 0$$

- In a circuit, steady current flows in wires.
- Consider a "node" in a circuit.
- We have for a node in a circuit

$$\boxed{\sum I = 0}$$



Kirchoff's Current Law

- Strictly speaking KCL only applies to circuits with steady currents (DC).
- However, for AC circuits having dimensions much smaller than a wavelength, KCL is also approximately applicable.

Question

- We have stated in this lecture that ‘strictly speaking, KCL applies only to DC’ and in a previous lecture that ‘strictly speaking, KVL applies only to DC’. What specific physical mechanisms are we neglecting when we apply KCL / KVL to AC circuits?

Summary

- The charge-current continuity equation arises from the fact that charges can be neither created nor destroyed.
- In the static case, the current density vector is solenoidal which leads to KCL.
- KCL states that the sum of currents entering a node in a circuit is zero.

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